

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Mock Interview

COURSE NAME: COMPUTER VISION WITH OPENCV
COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Understand video processing - motion computation - 3D vision - geometry. (BTL 6)

Assessment Tool Weightage: 10%

Mock Interview Questions

- 1. Design and explain a complete OpenCV pipeline to read, process, and display multiple images from a folder efficiently.**

Answer:

An efficient OpenCV image-processing pipeline can be designed in the following steps:

Step 1: Import required libraries

Use OpenCV (cv2) and Python's os module to access and process images.

Step 2: Read images from the folder

Get all image filenames from the folder and read them using cv2.imread().

Step 3: Validate the image

Check whether the image was loaded successfully. If the image is corrupted or unavailable, skip it instead of stopping the entire program.

Step 4: Process the image

Perform required operations such as resizing, grayscale conversion, filtering, or edge detection.

Step 5: Display the image

Use cv2.imshow() to display the processed images.

Step 6: Release resources

Use cv2.waitKey() and cv2.destroyAllWindows() to properly handle the display windows.

Pipeline:

Folder → Read Images → Validate → Preprocess → Process → Display → Release Resources

Optimizations:

- Process only supported image formats.
- Resize large images to reduce computation.
- Avoid loading all images into memory simultaneously.
- Handle corrupted files using error checking.
- Use modular functions for better maintainability.

This pipeline is efficient because each image is processed independently and unnecessary memory usage is avoided.

2. A video file is not loading properly in OpenCV. Analyze the possible causes and propose a structured debugging approach.

Answer:

A video may fail to load in OpenCV due to several reasons such as an incorrect file path, unsupported video format, missing codec, corrupted video, or incorrect use of VideoCapture().

Structured debugging approach:

Step 1: Check the file path

First, verify that the video file exists at the specified location and that the path is correct.

Step 2: Open the video using OpenCV

```
cap = cv2.VideoCapture("video.mp4")
```

Step 3: Check whether the video opened successfully

```
if not cap.isOpened():
```

```
    print("Unable to open video")
```

If `isOpened()` returns `False`, check the file format, codec, and path.

Step 4: Check frame reading

```
ret, frame = cap.read()
```

If `ret` is `False`, OpenCV is unable to read a frame.

Step 5: Check video properties

Check properties such as frame width, height, frame rate, and number of frames. Invalid or zero values can indicate a problem with the video or codec.

Step 6: Test another video

Try opening another standard video file. If the second video works, the original video may be corrupted or encoded using an unsupported codec.

Step 7: Release resources

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Common causes:

- Wrong file path
- Corrupted video file
- Unsupported format or codec
- Missing codec support
- Incorrect OpenCV installation
- Permission issues

Thus, debugging should proceed systematically from **file path** → **video opening** → **frame reading** → **video properties** → **codec/file verification**.

3. You need to overlay text and shapes on a live video feed. Design your solution and explain how you would implement it efficiently.

Answer:

To overlay text and shapes on a live video feed, OpenCV can capture frames continuously from a webcam and draw the required elements on each frame.

Pipeline:

Webcam → Capture Frame → Process Frame → Draw Text/Shapes → Display → Repeat

Implementation:

Step 1: Capture the webcam feed

```
cap = cv2.VideoCapture(0)
```

Step 2: Read frames continuously

```
ret, frame = cap.read()
```

Step 3: Draw shapes

OpenCV provides functions such as:

- `cv2.rectangle()` – draws rectangles
- `cv2.circle()` – draws circles
- `cv2.line()` – draws lines

Step 4: Add text

Use `cv2.putText()` to display information such as labels, status, or frame number.

Example:

```
cv2.putText(frame, "Live Feed",  
            (20, 40),  
            cv2.FONT_HERSHEY_SIMPLEX,  
            1, (255,255,255), 2)
```

Step 5: Display the processed frame

```
cv2.imshow("Video", frame)
```

Step 6: Exit and release resources

```
if cv2.waitKey(1) & 0xFF == ord('q'):
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Efficiency considerations:

- Avoid unnecessary processing on every frame.
- Keep drawing operations simple.
- Resize frames when high resolution is not required.
- Process only the required region of the frame when possible.
- Use cv2.waitKey(1) for smooth real-time display.

This approach provides a simple and efficient way to add **real-time annotations, labels, bounding boxes, and other graphical information** to a live video feed.

Code of Conduct:

*I V G ABBIRAMY - 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

V G ABBIRAMY

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledg e	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides wellstructured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communic ation	15	Communicate s clearly, confidently,	Communicate s well with minor	Communicatio n lacks clarity or confidence	Unable to communicate effectively

		and professionally with proper terminology	hesitation		
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Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately